UNIT - III
Limit State Design of Slabs

Design of Reinforced Concrete Structures (23A01502) - handwritten study notes

1. Introduction and Classification of Slabs

- A **slab** is a plane structural element, thin in comparison with its other two dimensions. It carries transverse loads (floor, roof, live load) mainly by **bending** and transfers them to beams, walls or columns.
- Design is done by the **Limit State Method** of **IS 456:2000**. Concrete M20 and steel Fe415 are used in the examples below.
- Slabs are classified by **support condition** and by **direction of load transfer**.

One-way slab

- Supported on two opposite sides only, **or** supported on four sides with ratio $l_y/l_x > 2$.
- Main steel is provided in the **short span** direction. Distribution (temperature) steel is provided in the long direction.
- Slab bends in **one direction** - designed as a 1 m wide beam strip.

Two-way slab

- Supported on all four sides with $l_y/l_x <= 2$. Bending takes place in **both directions**.
- Main steel is provided in both directions; the short span carries a larger share of the load.

Rule: $l_y / l_x > 2 \longrightarrow$ One-way slab. $l_y / l_x <= 2 \longrightarrow$ Two-way slab. Here l_x = short span, l_y = long span.

Feature	One-way slab	Two-way slab
Supports	Two opposite sides (or ratio > 2)	Four sides, ratio <= 2

Bending	In short direction only	In both directions
Main steel	One direction	Both directions
Distribution steel	Required in long direction	Not required (main steel both ways)
Torsion steel at corners	Not needed	Needed at discontinuous corners

Loads on slabs

- **Dead load**: self weight = D x 25 kN/m^3, floor finish (about 1.0 kN/m^2), partitions.
- **Imposed (live) load** as per IS 875 Part 2: residences 2 kN/m^2, offices 2.5 - 4 kN/m^2, balconies 3 kN/m^2, classrooms 3 kN/m^2.
- **Factored load** $w_u = 1.5\,(DL + LL)$.

2. Basic Design Data (IS 456:2000)

Effective span (Cl. 22.2)

- **Simply supported slab**: the smaller of (i) clear span + effective depth d, (ii) centre to centre of supports.
- **Cantilever**: length to the face of support + d/2 (cantilever which is the end of a continuous slab: to the centre of support).
- **Continuous slab**: centre to centre of supports (if support width is small compared to span).

Minimum depth: span / effective depth ratio (Cl. 23.2.1)

Support condition	Basic l/d
Cantilever	7
Simply supported	20
Continuous	26

- Basic value is **multiplied by a modification factor** k_t from Fig. 4 of IS 456, which depends on the percentage steel p_t and the service stress f_s.

- Service stress $f_s = 0.58\, f_y \times (A_{st\ required} / A_{st\ provided})$.
- For two-way slabs up to 3.5 m span with live load up to 3 kN/m²: **simply supported 35**, **continuous 40** for mild steel; multiply by **0.8** for Fe415 (28 and 32).
- For spans over 10 m the values are multiplied by 10 / span (not for cantilevers).

Flexure formulae (singly reinforced, per 1 m width)

$$M_{u,lim} = 0.138\, f_{ck}\, b\, d^2 \quad (Fe415)$$

$$M_u = 0.87\, f_y\, A_{st}\, d\, [1 - A_{st}\, f_y / (b\, d\, f_{ck})]$$

$$A_{st} = (0.5\, f_{ck} / f_y)\, [1 - \mathrm{sqrt}(1 - 4.6\, M_u / (f_{ck}\, b\, d^2))]\, b\, d$$

- Slabs are always designed **under-reinforced** ($x_u < x_{u,max}$); doubly reinforced slabs are not normally needed.
- Overall depth D = d + clear cover + half the bar diameter. Minimum **clear cover 20 mm** (mild exposure); 15 mm may be used for D < 100 mm in some cases.

Reinforcement limits

- **Minimum steel** (Cl. 26.5.2.1): 0.15 % of b D for mild steel (Fe250); **0.12 %** of b D for HYSD bars (Fe415, Fe500).
- **Maximum bar diameter** <= D / 8.
- **Spacing of main steel**: not more than **3d** or **300 mm**, whichever is smaller.
- **Spacing of distribution steel**: not more than **5d** or **450 mm**, whichever is smaller.
- Minimum bar diameter used is generally 8 mm for main steel, 6 or 8 mm for distribution steel.

> Remember: main bars ⟶ 3d / 300 mm. Distribution bars ⟶ 5d / 450 mm. Minimum steel ⟶ 0.12 % (HYSD) or 0.15 % (mild steel) of bD.

Shear in slabs (Cl. 40.2)

- Nominal shear stress: $\tau_v = V_u / (b\ d)$.
- Slabs are generally safe in shear; no shear reinforcement is provided, the slab is safe if $\tau_v < k\ \tau_c$.
- τ_c is taken from **Table 19** (depends on p_t and grade of concrete). For M20: p_t 0.15 ⟶ 0.28, 0.25 ⟶ 0.36, 0.50 ⟶ 0.48, 0.75 ⟶ 0.56, 1.0 ⟶ 0.62 N/mm^2.
- The factor **k** depends on overall depth (Cl. 40.2.1.1):

Overall depth D (mm)	300 or more	275	250	225	200	175	150 or less
k	1.00	1.05	1.10	1.15	1.20	1.25	1.30

- Maximum shear stress $\tau_{c,max}$ for M20 = 2.8 N/mm^2; if exceeded, increase depth.

3. Design of Cantilever Slab

- Cantilever slabs have **hogging moment**; main steel is placed at the **top** (tension face) and extends well into the backing slab/beam.
- **Design moment**: $M_u = w_u L^2 / 2$, shear $V_u = w_u L$ (at the support).
- The slab is usually tapered (thicker at the support, thinner at the free end): depth at the support = L / 7 x k_t approximately.
- Main steel should be **anchored** into the support by at least L_d and preferably continued into the adjacent span.
- Distribution steel is placed under the main steel (at right angles) at 0.12 % of bD.

Assume depth d = L / (7 k_t) → Load w_u (incl. parapet) → $M_u = w_u L^2/2$ → A_{st} at top

Distribution steel → Check shear and deflection

Example 1: Cantilever slab (balcony)

Problem: Design a cantilever balcony slab of projection 1.5 m. Live load 3 kN/m^2, floor finish 1 kN/m^2. Use M20 and Fe415.

- **Depth**: take D = 180 mm, d = 180 - 20 - 5 = 155 mm. Check: L/d = 1500 / 155 = 9.7 < 7 x k_t (with k_t around 1.8, allowable 12.6).
- **Loads**: self weight 0.18 x 25 = 4.5, finish 1.0, live 3.0. Total = 8.5 kN/m^2. **w_u = 1.5 x 8.5 = 12.75 kN/m**.
- **M_u** = 12.75 x 1.5^2 / 2 = **14.34 kNm**; V_u = 12.75 x 1.5 = 19.1 kN.
- Check depth: d = sqrt(14.34 x 10^6 / (0.138 x 20 x 1000)) = 72 mm < 155 mm. So the section is under-reinforced.
- **Steel**: M_u / (b d^2) = 14.34 x 10^6 / (1000 x 155^2) = 0.597. A_{st} = 0.024 x [1 - sqrt(1 - 4.6 x 0.597 / 20)] x 1000 x 155 = **266 mm^2**.
- Use 10 mm bars: spacing = 78.5 x 1000 / 266 = 295 mm. Provide **10 mm @ 275 mm c/c** (A_{st} = 285 mm^2) at top. Spacing < 3d = 465 mm. OK.
- **Distribution steel** = 0.12 % x 1000 x 180 = 216 mm^2. Provide **8 mm @ 230 mm c/c** (219 mm^2).
- **Shear**: tau_v = 19100 / (1000 x 155) = 0.123 N/mm^2. p_t = 100 x 285 / (1000 x 155) = 0.184 %, tau_c = 0.30. k = 1.24 ⟶ k tau_c = 0.37 > 0.123. **Safe**.
- **Deflection**: f_s = 0.58 x 415 x (266/285) = 225 N/mm^2; k_t about 1.8. Allowable l/d = 7 x 1.8 = 12.6 > 9.7. **Safe**.

Ans: D = 180 mm; main steel 10 mm @ 275 c/c on top, extended into the support by L_d; distribution 8 mm @ 230 c/c.

4. One-Way Simply Supported Slab

- Design a **1 m wide strip** as a simply supported beam of span l_{eff}.

$$M_u = w_u l^2 / 8 \quad V_u = w_u l / 2$$

- The reinforcement bars are placed at the **bottom** and **spaced** such that spacing <= 3d and 300 mm.
- **Alternate bars** may be bent up near the support at about l/7 from the face (for continuity and anchorage); at least **one third** of the positive steel must be extended into the support (Cl. 26.2.3.3).
- At a simple support the bars must be anchored: $M_1 / V + L_0 >= L_d$, where L_0 = embedment beyond centre of support (not more than d or 12 diameters).
- **Development length** $L_d = 0.87 f_y \text{ phi} / (4 \tau_{bd})$. For M20 and Fe415 ($\tau_{bd}$ = 1.2 x 1.6 = 1.92 N/mm²): **L_d = 47 phi**.

Example 2: One-way simply supported slab

Problem: Design a one-way slab for a room with clear span 3.5 m, walls 230 mm thick on both sides. Live load 4 kN/m², floor finish 1 kN/m². M20, Fe415.

- **Depth**: $l/d = 20 \times k_t$; assume $k_t = 1.4 \longrightarrow d = 3600 / 28 = 129$ mm. Provide D = 170 mm, d = 170 - 25 - 5 = **140 mm** (12 mm bars, 20 mm cover).
- **Effective span**: clear + d = 3.5 + 0.14 = 3.64 m; c/c = 3.5 + 0.23 = 3.73 m. **l_{eff} = 3.64 m**.
- **Loads**: self weight 0.17 x 25 = 4.25, finish 1.0, live 4.0 $\longrightarrow$ 9.25 kN/m². **w_u = 1.5 x 9.25 = 13.875 kN/m**.
- **M_u** = 13.875 x 3.64² / 8 = **22.98 kNm**; **V_u** = 13.875 x 3.64 / 2 = **25.25 kN**.
- Depth check: $d_{min} = \text{sqrt}(22.98 \times 10^6 / 2760) = 91$ mm < 140 mm. OK.
- **Steel**: $M_u / (b d^2) = 22.98 \times 10^6 / (1000 \times 140^2) = 1.172$. A_{st} = 0.024096 x [1 - sqrt(1 - 4.6 x 1.172 / 20)] x 1000 x 140 = **490 mm²**.

- 12 mm bars: spacing = 113 x 1000 / 490 = 230 mm. Provide **12 mm @ 230 mm c/c** (A_{st} = 491 mm^2). 230 < 3d = 420 and < 300. OK.
- **Distribution steel** = 0.12 % x 1000 x 170 = 204 mm^2: **8 mm @ 240 mm c/c** (209 mm^2) < 5d = 700 and 450. OK.
- **Shear**: tau_v = 25250 / (1000 x 140) = 0.18 N/mm^2; p_t = 0.35 %, tau_c = 0.40; k = 1.25 ⟶ k tau_c = 0.50 > 0.18. **Safe**.
- **Deflection**: f_s = 0.58 x 415 x 490/491 = 240; p_t = 0.35 ⟶ k_t about 1.4 (Fig. 4). Allowable l/d = 20 x 1.4 = 28; actual = 3640 / 140 = 26. **Safe**.
- **Development length**: L_d = 47 x 12 = 564 mm. At support: M_1 / V = 0.87 x 415 x 491 x 140 / 25250 = 983 mm, so anchorage is OK.

Ans: D = 170 mm; main steel 12 mm @ 230 c/c; distribution 8 mm @ 240 c/c; bars bent up alternately near the support.

5. Continuous Slabs

- Continuous slabs are economical: smaller moments and smaller depth than simply supported slabs. Spans may be taken as c/c of supports.
- **IS 456 Table 12** gives bending moment coefficients and **Table 13** gives shear force coefficients for continuous beams and slabs of **three or more equal spans** (or spans differing by not more than 15 %), carrying uniformly distributed loads, where the imposed load does not exceed 3/4 of the dead load (Cl. 22.5.1).

Table 12: Bending moment = coefficient x w l^{2}

Type of load	Mid of end span	Support next to end support	Mid of interior span	Other interior supports
Dead load (fixed)	+1/12	-1/10	+1/16	-1/12
Imposed load (not fixed)	+1/10	-1/9	+1/12	-1/9

Table 13: Shear force = coefficient x w l

Type of load	End support	Next to end - outer side	Next to end - inner side	Other supports
Dead load (fixed)	0.4	0.6	0.55	0.5
Imposed load (not fixed)	0.45	0.6	0.6	0.6

- Moment for each load case = (coefficient for DL) x $w_d l^2$ + (coefficient for LL) x $w_l l^2$, where w_d and w_l are the **factored** dead and imposed loads.
- The effective span l is the **centre to centre** distance; for two spans only, or unequal spans, moment distribution should be done.
- Design the section for the **maximum moment** (usually the support next to the end support); use the same depth throughout.
- **Detailing**: positive steel at the bottom in the span, negative steel on top over the supports; **bent-up bars** in spans are carried over the support (alternate bars bent up and extended as top steel).

Example 3 (outline): Continuous slab with three equal spans of 3.5 m

- Take D = 130 mm, d = 105 mm. Dead load = 0.13 x 25 + 1.0 = 4.25 kN/m^2; live load = 4 kN/m^2.
- Factored w_d = 1.5 x 4.25 = 6.375 kN/m; w_l = 1.5 x 4 = 6.0 kN/m (per metre width). l^2 = 12.25.

Section	Dead load part	Imposed load part	Total M_u (kNm)
Mid of end span	6.375 x 12.25 / 12 = 6.51	6.0 x 12.25 / 10 = 7.35	+13.86
Support next to end support	6.375 x 12.25 / 10 = 7.81	6.0 x 12.25 / 9 = 8.17	-15.98
Mid of interior span	6.375 x 12.25 / 16 = 4.88	6.0 x 12.25 / 12 = 6.13	+11.01

Other interior supports	6.375 x 12.25 / 12 = 6.51	6.0 x 12.25 / 9 = 8.17	-14.68

- **Maximum shear** (support next to end, outer side): 0.6 x (6.375 + 6.0) x 3.5 = **26.0 kN**.
- Depth check: d_{min} = sqrt(15.98 x 10^6 / 2760) = 76 mm < 105 mm. OK.
- A_{st} at the support next to end support = **464 mm²** (M_u / b d^2 = 1.449) ⟶ 10 mm @ 160 mm c/c (491 mm^2) top steel. Other sections are designed in the same way. Distribution steel 0.12 % x 1000 x 130 = 156 mm^2 ⟶ 8 mm @ 300 c/c.

Remember: for continuous slabs read coefficients separately for dead and imposed load; hogging moments are larger than sagging moments.

6. Two-Way Slabs - IS 456 Annex D

- Two-way slabs are designed by the **coefficient method** of **Annex D** (Cl. 24.4).
- **Annex D-1**: simply supported slabs with corners **free to lift** - Table 27.
- **Annex D-2**: slabs **restrained** at the edges (corners held down) - Table 26.

$$M_x = \text{alpha}_x\, w_u\, l_x^2 \quad M_y = \text{alpha}_y\, w_u\, l_x^2$$

- l_x = shorter effective span; l_y = longer effective span; M_x acts on the strip of span l_x, M_y on the strip of span l_y. Both moments are per metre width.
- alpha_x, alpha_y are moment coefficients that depend on l_y / l_x and the **edge condition**.

Nine edge conditions in Table 26

- **Case 1** interior panel; **2** one short edge discontinuous; **3** one long edge discontinuous; **4** two adjacent edges discontinuous.

- **Case 5** two short edges discontinuous; **6** two long edges discontinuous; **7** three edges discontinuous (one long edge continuous); **8** three edges discontinuous (one short edge continuous); **9** four edges discontinuous.

Selected coefficients (Table 26) for alpha_{x}

Case	Moment	l_y/l_x = 1.0	1.5	2.0
1 Interior	Negative	0.032	0.053	0.065
1 Interior	Positive	0.024	0.041	0.049
4 Two adjacent edges disc.	Negative	0.047	0.075	0.091
4 Two adjacent edges disc.	Positive	0.035	0.056	0.069
9 Four edges disc.	Positive	0.056	0.089	0.107

- For case 1: **$alpha_y$** = 0.032 (negative) and 0.024 (positive) for all values of l_y/l_x. For case 4 $alpha_y$ = 0.047 (negative) and 0.035 (positive).
- Read other values from **Table 26** of IS 456 and interpolate for intermediate ratios.

Table 27: Simply supported slab, corners not held down

l_y/l_x	1.0	1.1	1.2	1.3	1.4	1.5	1.75	2.0
$alpha_x$	0.062	0.074	0.084	0.093	0.099	0.104	0.113	0.118
$alpha_y$	0.062	0.061	0.059	0.055	0.051	0.046	0.037	0.029

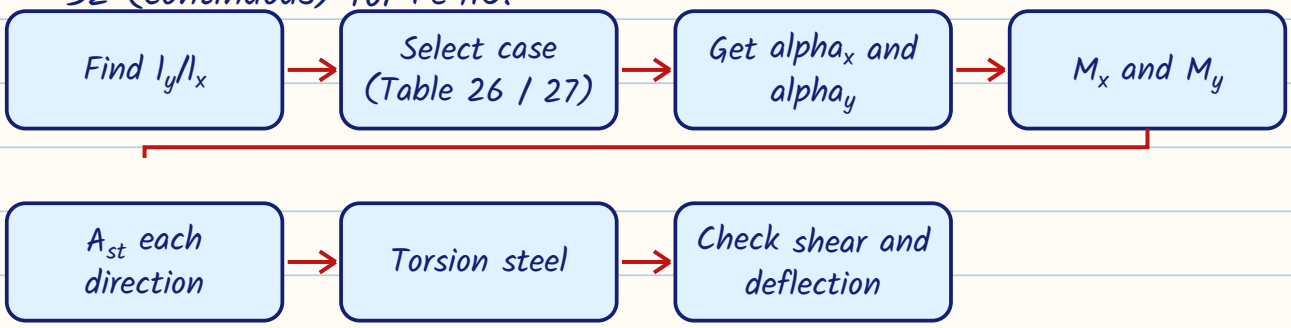
Torsion reinforcement at corners (Cl. D-1.8)

- When corners are **prevented from lifting**, torsion steel is provided at the corners where the slab is discontinuous.
- **Corner with both edges discontinuous**: area of steel in each of the four layers (top and bottom, in both directions) = **3/4 of the area required for the maximum mid-span moment** (short span).
- **Corner with one edge continuous and the other discontinuous**: **half** of the above (3/8 of the mid-span area).
- **No torsion steel** where the slab is continuous over both edges.
- The torsion steel extends from the edge a distance of **l_x / 5** in each

direction.

Other design rules for two-way slabs

- The slab is divided into a **middle strip** (3/4 of the width) and **two edge strips** (each 1/8 of the width) in each direction. The middle strip carries the full moments; edge strips need only **minimum steel** (0.12 %).
- Short-span bars are placed **nearer to the tension face** (outer layer, larger d); long-span bars lie above them with d reduced by one bar diameter: $d_y = d_x - phi$.
- **Discontinuous edges**: negative moment at the discontinuous edge = 1/3 of the mid-span moment (or as given in the table).
- Bar spacing limits: 3d / 300 mm for main steel in both directions.
- **Shear**: $V_u = w_u l_x / 2$ per metre width (check $tau_v < k\ tau_c$); the load is shared between the two directions, so the short-span value is conservative.
- **Deflection**: use l_x / d with the basic values 28 (simply supported) and 32 (continuous) for Fe415.



Remember: two-way slab ⟶ short span steel is the bottom layer; torsion steel = 3/4 of mid span steel (two discontinuous edges) and 1/2 of that for one discontinuous edge; length $l_x/5$.

Example 4: Restrained two-way slab (corners held down)

Problem: A slab panel 4 m x 5 m (clear spans) is discontinuous along two adjacent edges and continuous on the other two edges. Live load 3 kN/m^2, floor finish 1 kN/m^2, M20, Fe415. Design the slab.

- $l_y / l_x = 5 / 4 = 1.25 \longrightarrow$ Case 4 (two adjacent edges discontinuous). Coefficients (interpolated for 1.25): $alpha_x(-) = 0.0625$, $alpha_x(+) = 0.047$, $alpha_y(-) = 0.047$, $alpha_y(+) = 0.035$.
- **Depth**: l/d = 28 (Fe415, simply supported) x k_t > 4000 / 28 = 143 mm; for continuous slabs a smaller depth can be accepted. Take **D = 150 mm**, d_x = 125 mm, d_y = 115 mm (10 mm bars).
- **Load**: self weight 0.15 x 25 = 3.75 + 1.0 + 3.0 = 7.75 kN/m^2; **w_u = 11.625 kN/m^2**. $w_u l_x^2$ = 11.625 x 16 = 186 kNm.

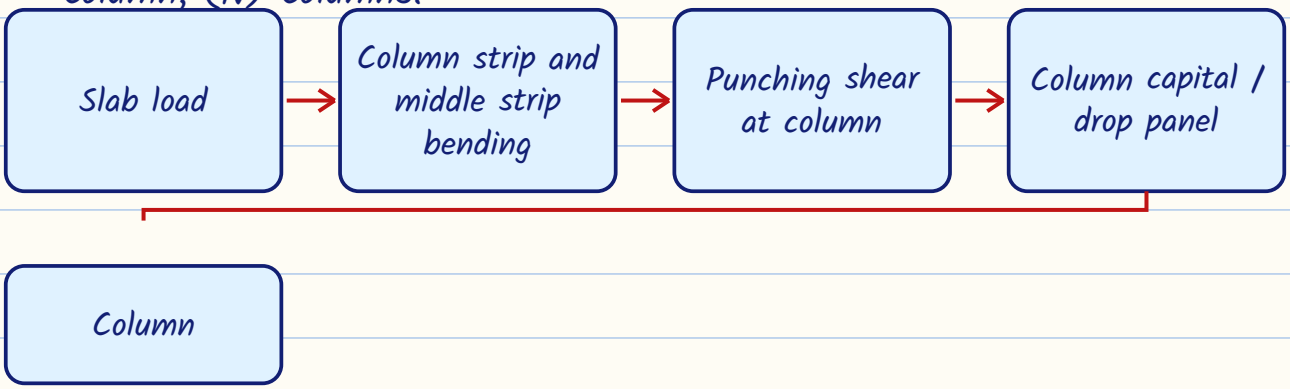
Moment	Coefficient	M_u (kNm/m)	d (mm)	A_{st} req. (mm^2)	Provide
M_x negative	0.0625	11.63	125	270	10 mm @ 280
M_x positive	0.047	8.74	125	200	8 mm @ 240
M_y negative	0.047	8.74	115	219	8 mm @ 220
M_y positive	0.035	6.51	115	162 (min 180)	8 mm @ 270

- Minimum steel = 0.12 % x 1000 x 150 = **180 mm^2**. Spacing < 3d = 375 mm and < 300 mm in all cases.
- **Torsion steel** at the corner where two adjacent edges are discontinuous: 3/4 x 200 = **150 mm^2** in each of four layers $\longrightarrow$ 8 mm @ 300 mm c/c, extended over l_x/5 = 4000 / 5 = **800 mm** from the edges. At corners with one discontinuous edge: half of this.
- **Shear**: V_u = 11.625 x 4 / 2 = 23.25 kN/m; tau_v = 23250 / (1000 x 125) = 0.186 N/mm^2. p_t = 100 x 270 / (1000 x 125) = 0.22 %, tau_c about 0.33, k = 1.3 $\longrightarrow$ 0.43 > 0.186. **Safe**.
- **Deflection**: l/d = 4000 / 125 = 32; allowable = 28 x k_t (k_t about 1.3 $\longrightarrow$ 36). **Safe**.

8. Introduction to Flat Slabs

- A **flat slab** is a reinforced concrete slab supported **directly on columns** without beams. The loads are transferred to the columns by bending and shear (punching) in the slab.

- **Components**: (i) slab, (ii) **drop panel** - thickened part of the slab around the column, (iii) **column capital** (head) - flared top of the column, (iv) columns.



Drop panel and column capital (IS 456 Cl. 31)

- **Drop panel**: should extend in each direction at least **1/3 of the panel length** in that direction; its projection below the slab should not be more than **D/4**; the total depth used in the design near the column is limited to slab thickness + D/4.
- **Column capital**: is designed for the effective diameter measured where the 45 degree line from the capital meets the soffit of the slab (or drop). The usual dimension is about 0.25 l for the capital diameter.
- **Minimum thickness** of a flat slab = **125 mm**. Span/depth ratio: l/d as per Cl. 23.2 (for flat slabs the longer span is used; multiply by 0.9 when there is no drop panel).

Column strip and middle strip

- **Column strip**: width on each side of the column centreline = 0.25 x the **smaller** of l_1 and l_2.
- **Middle strip**: the remaining part of the panel between two column strips.
- Moments are first found for the whole panel and then divided between column strip and middle strip.

Direct design method (Cl. 31.4)

- Conditions: at least **three continuous spans** in each direction; ratio of longer to shorter span not more than 2; successive span lengths differ

by not more than 1/3 of the longer span; column offset not more than 10 % of span; **live load not more than 3 times the dead load**; loads uniformly distributed.

$$M_o = W l_n / 8 \quad W = w_u l_2 l_n$$

- l_n = clear span in the direction of moments, l_2 = span transverse to l_1.
- **Interior panel**: negative design moment = **0.65 M_o**, positive design moment = **0.35 M_o**.

Moment	Column strip (% of panel moment)	Middle strip
Negative (at support)	75 %	25 %
Positive (mid span)	60 %	40 %

Punching (two-way) shear check (Cl. 31.6)

- The critical section is at a distance **d/2 from the periphery of the column / capital / drop**.
- Perimeter $b_o = 2 (c_1 + d) + 2 (c_2 + d)$ for a rectangular column of sides $c_1 \times c_2$.

$$tau_v = V_u / (b_o d) <= k_s tau_c$$

- tau_c = **0.25 sqrt(f_{ck})** (= 1.12 N/mm^2 for M20) and **k_s = 0.5 + $beta_c$** (not more than 1), where $beta_c$ = ratio of short side to long side of the column.
- If the check fails: increase thickness, provide a **drop panel**, a larger column capital, or shear reinforcement (stirrups / shear heads).

Advantages and limitations of flat slabs

Advantages	Limitations
Flat soffit - simple formwork	Higher punching shear risk at columns
Less storey height / better headroom	Thicker slab (more concrete) for the same span

Remember: drop panel >= l/3, projection <= D/4, punching at d/2 from column, tau_c = 0.25 sqrt(f_{ck}), min thickness 125 mm.

Quick Revision

- One-way: l_y/l_x > 2; two-way: <= 2. Main steel in short span; distribution 0.12 %.
- Basic l/d: **7 (cantilever), 20 (simply supported), 26 (continuous)** x k_t.
- Shear check: tau_v < k tau_c; k = 1.3 for D <= 150 down to 1.0 for D >= 300.
- Continuous slab: Table 12 (moments) and Table 13 (shears); live load <= 3/4 dead load.
- Two-way: M_x = $alpha_x$ w l_x^2, M_y = $alpha_y$ w l_x^2; Table 26 (restrained) and Table 27 (simply supported).
- Torsion steel: 3/4 of mid span (two discontinuous edges), 1/2 (one), none (continuous); length l_x/5.
- Flat slab: drop panel l/3, capital, column strip 0.25 l, M_o = W l_n/8, 0.65 / 0.35 split, punching at d/2.

Important Questions

- 1. Design a cantilever slab of 1.2 m projection for a live load of 3 kN/m^2. Use M20 and Fe415.
- 2. Design a one-way simply supported slab for a room 3 m x 8 m (clear), live load 3 kN/m^2, with detailing sketch.
- 3. State the coefficients of IS 456 Tables 12 and 13 for continuous slabs. Design an end span of a continuous slab.
- 4. Design a two-way slab 4 m x 5 m restrained at all edges with two adjacent edges discontinuous. Show the torsion reinforcement.
- 5. Write short notes on flat slabs: drop panel, column capital, column strip, punching shear and direct design method.